

Introduction

Predicting short-term atmospheric variability remains a central challenge in climate science and environmental studies. Recent advances in machine learning have opened new possibilities for extracting predictive patterns from large climate datasets. In this project, we investigate the use of artificial intelligence methods to forecast daily mean temperature using historical meteorological data. This approach provides a foundation for future studies aimed at understanding and predicting temperature extremes.

Objectives and Dataset

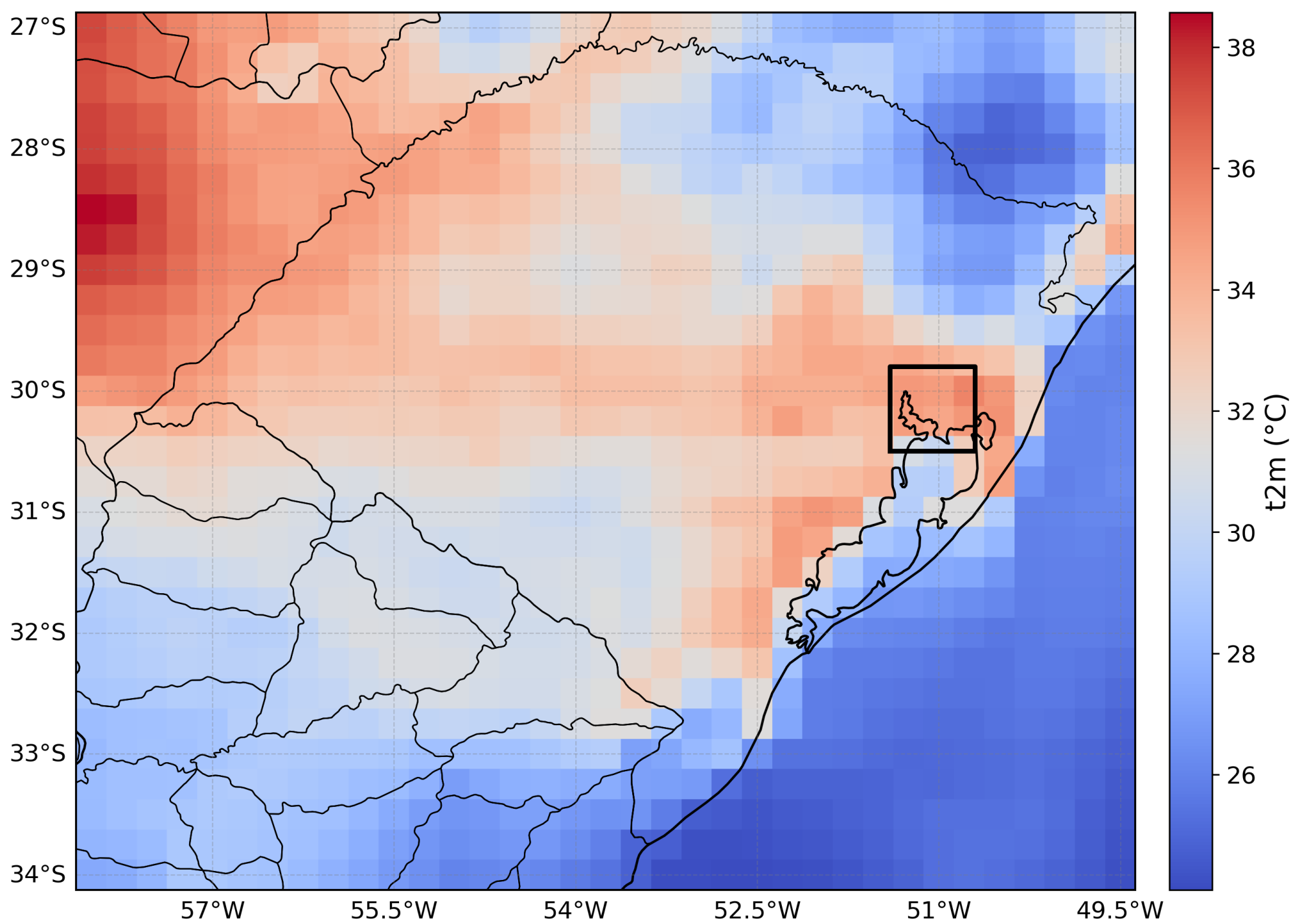
Develop a deep learning model capable of forecasting extreme weather events in Porto Alegre in advance from historical climate fields.

Specific objectives:

- Evaluate the predictive skill of temperature for lead times from 1 to 10 days ahead;
- Compare different temporal lags used as input sequences;
- Investigate the impact of spatial information on forecasting skill;

Dataset:

- Historical reanalysis / climate data by ERA5 from 1940-2025;
- Daily gridded fields over Southern Brazil in three different sizes (Poa1, RS and SES);
- Variables used: average temperature;
- Target region: Porto Alegre (4×4 grid box);
- Predictand: area daily average temperature.



Methodology

The original climate fields were preprocessed and standardized using z-score normalization. Input samples were built using sequences of past meteorological maps with temporal lag, allowing the network to learn spatial-temporal dependencies.

$$X' = \frac{X - \mu}{\sigma}$$

where μ is the mean and σ the standard deviation calculated on the training set.

Example of the entrance tensor:

$$(N, L, H, W)$$

where L = number of lags, $H \times W$ = spatial grid e N = number of samples.

The dataset was divided into training, validation, and testing subsets. The CNN was trained using the Adam optimizer and Huber loss function. Model performance was evaluated through regression metrics and visual comparison between observed and predicted time series.

The time period used for each subset was the 1990s, when climate change was not as pronounced as it is nowadays:

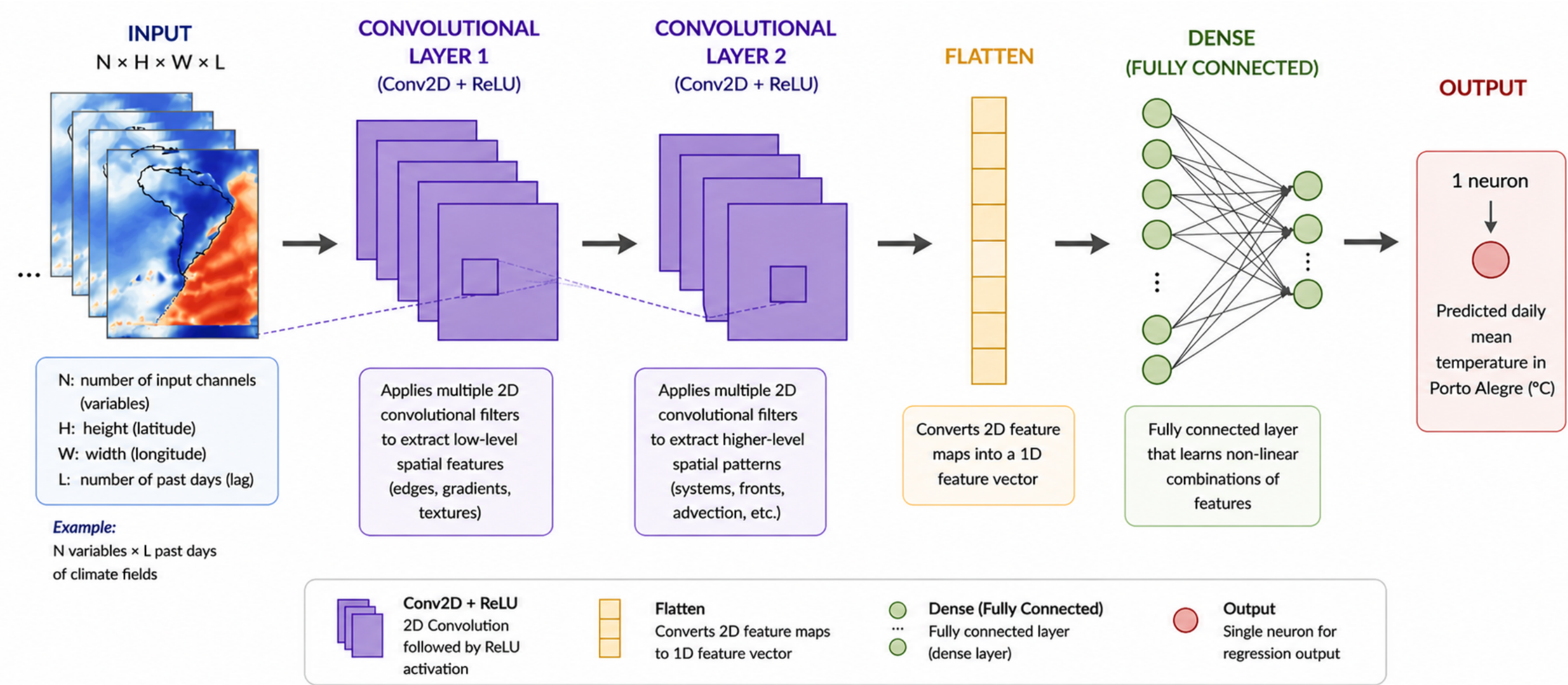
1. Training set: 1990-1996
2. Validation set: 1997
3. Test set: 1998-1999

Model Architecture

A two-dimensional CNN was implemented to process daily meteorological maps.

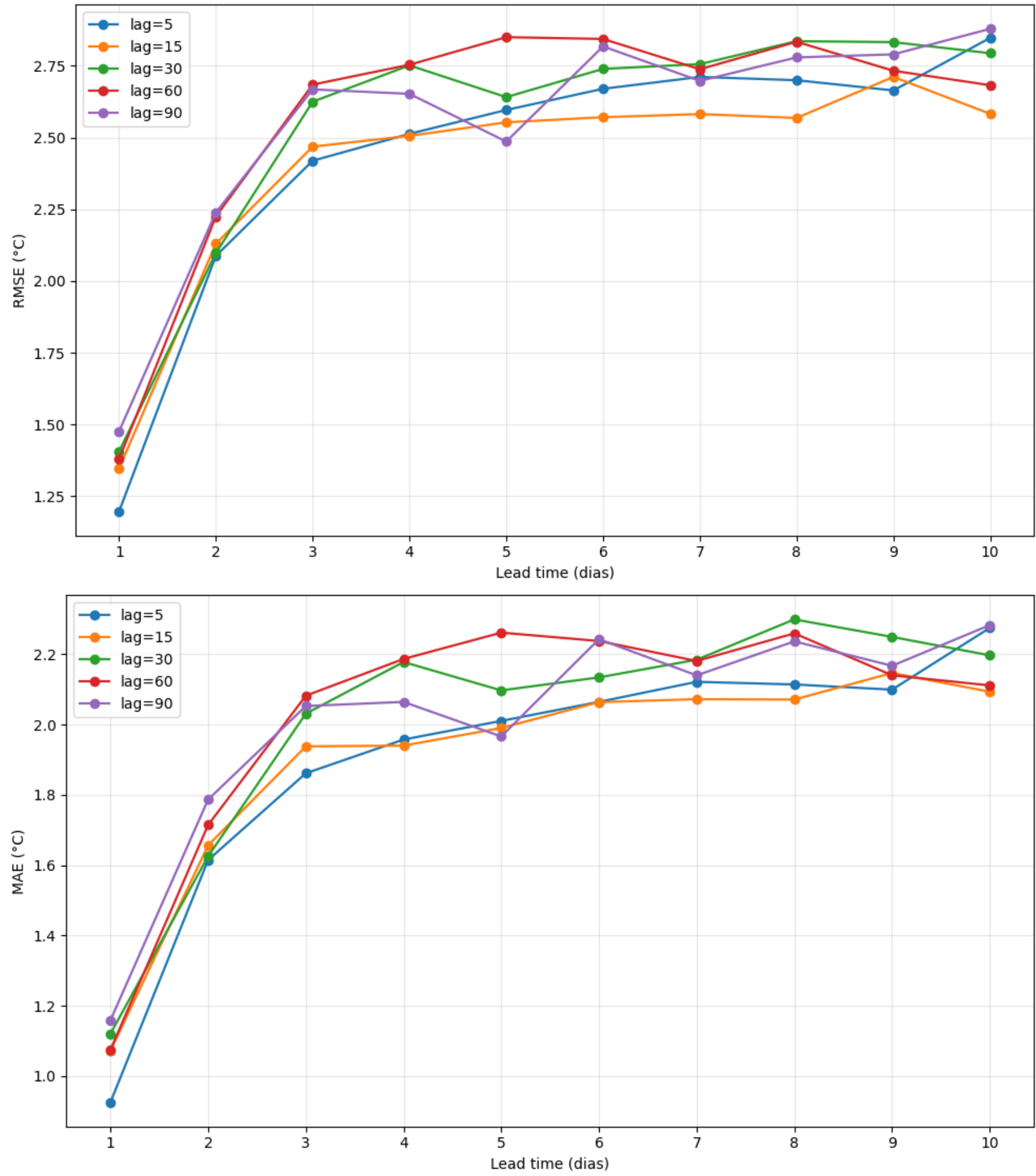
- Input: sequence of spatial temperature maps (only temperature information is used);
- Convolutional layers with ReLU activation;
- Max-pooling layers for dimensionality reduction;
- Flatten layer;
- Fully connected dense layers;
- Output neuron for temperature regression in the region of Porto Alegre.

The convolutional filters identify relevant regional structures related to heat transport and atmospheric persistence patterns.



Evaluation Metrics

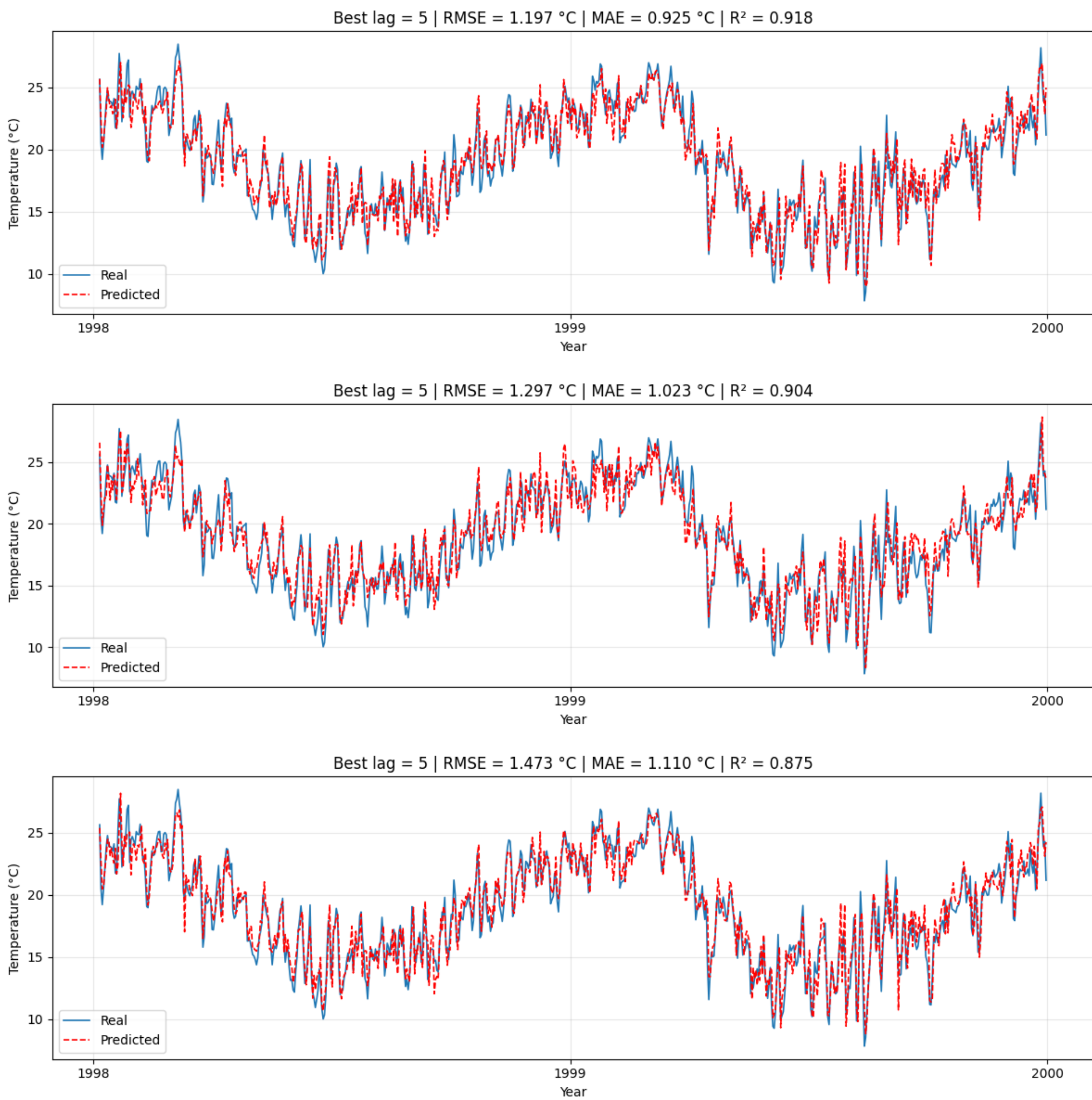
The metrics below (Root Mean Squared Error and Absolute Mean Error) show how the error in the forecast evolves with the lead time period used:



Results

The CNN model successfully reproduced the seasonal cycle and a large part of the day-to-day variability of mean temperature in Porto Alegre. Below are the result plots obtained using, respectively, the RS map, the SES map, and the Poa1 map, considering a lag of 5 days.

Time Series Forecast vs. Predicted Dispersion



These results indicate that CNNs can efficiently learn relevant spatial patterns associated with local temperature variability.

Conclusions and Future Perspectives

Conclusions:

- CNNs are effective tools for local climate prediction;
- Spatial information improves forecasting skill;
- Deep learning provides a computationally efficient alternative.

Future perspectives:

- Inclusion of exogenous variables (pressure, wind, geopotential);
- Use of hybrid CNN-LSTM architectures;
- Prediction of heatwaves and extreme events;
- Extension to other regions in Brazil;
- Probabilistic forecasting with uncertainty quantification.

References

- Trok, J. T. et al. ML-based extreme event attribution. Sci. Adv. (2024).
- Salcedo-Sanz, S. et al. Extreme atmospheric events with ML. Theor. Appl. Climatol. (2023).
- Mascolo, V. et al. Gaussian framework for extreme event prediction. JAMES (2025).
- WMO. Forecast Verification Research. Available at: <https://wmo.int/wwrpwgne-joint-working-group-forecast-verification-research>.